

VALVE POWER BOARD

RUBY - REV A



RESPONSIBLE ENGINEER:
NICHOLAS ALVES

REVIEWERS:

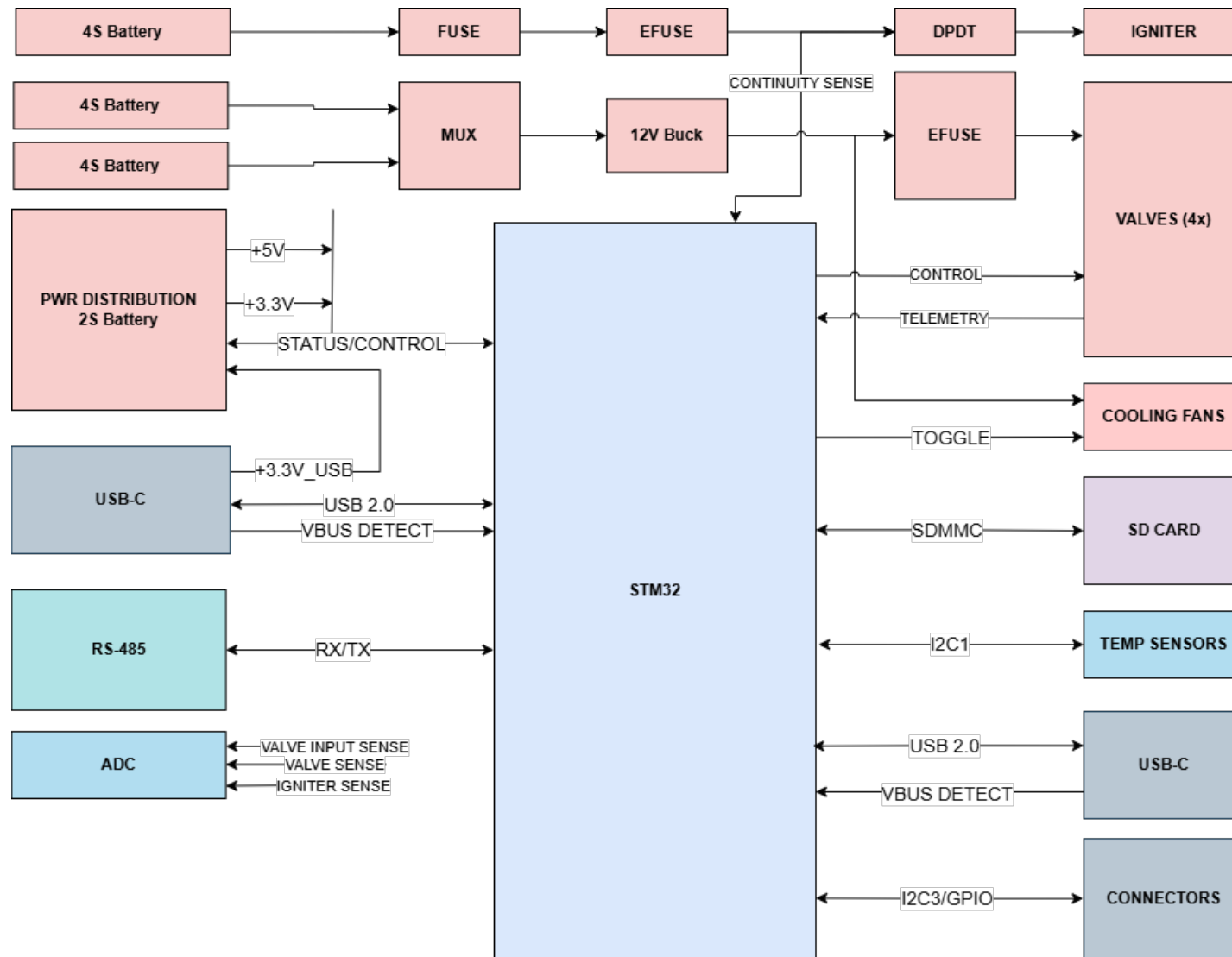
SCRT
STUDENT COMPETITION ROCKETRY TEAM

Sheet: /Title Page/
File: top-level.kicad_sch

Title: TITLE PAGE

Size: A4 Date: 2026-08-27
KiCad E.D.A. 10.0.5

Rev: A
Id: 1/32



Sheet: /Block Diagrams/
File: block_diagrams.kicad_sch

Title: BLOCK DIAGRAM

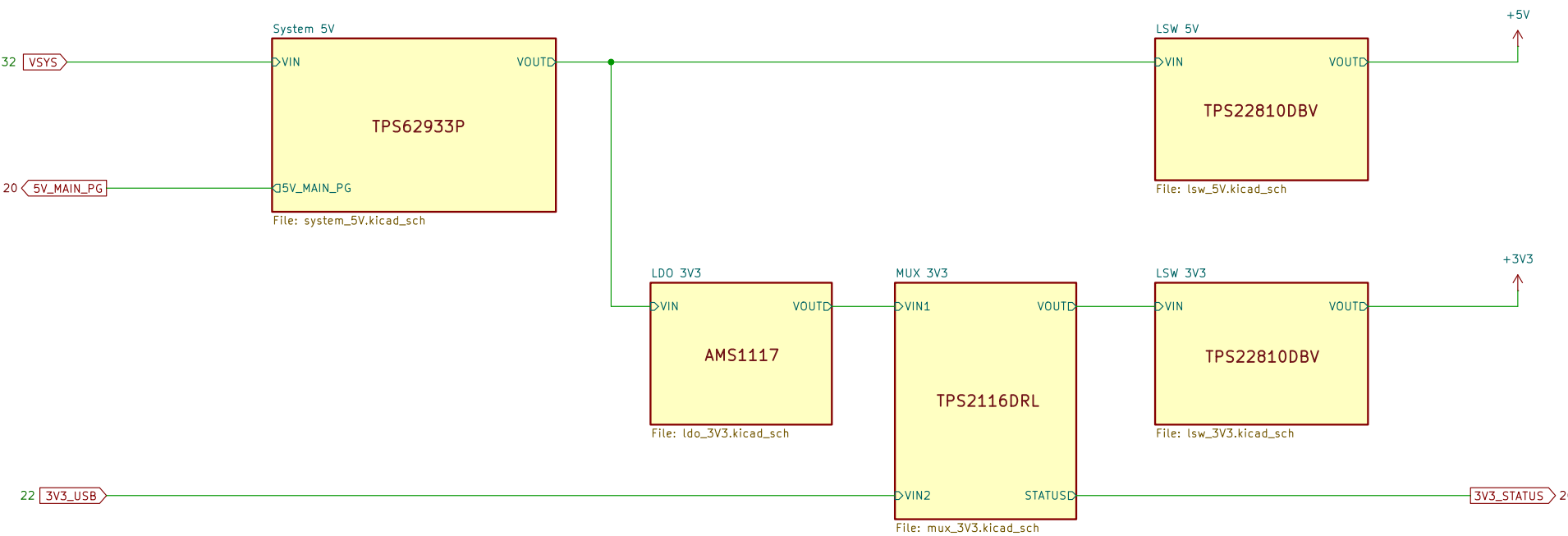
Size: A4 Date: 2026-08-27

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SYSTEM POWER



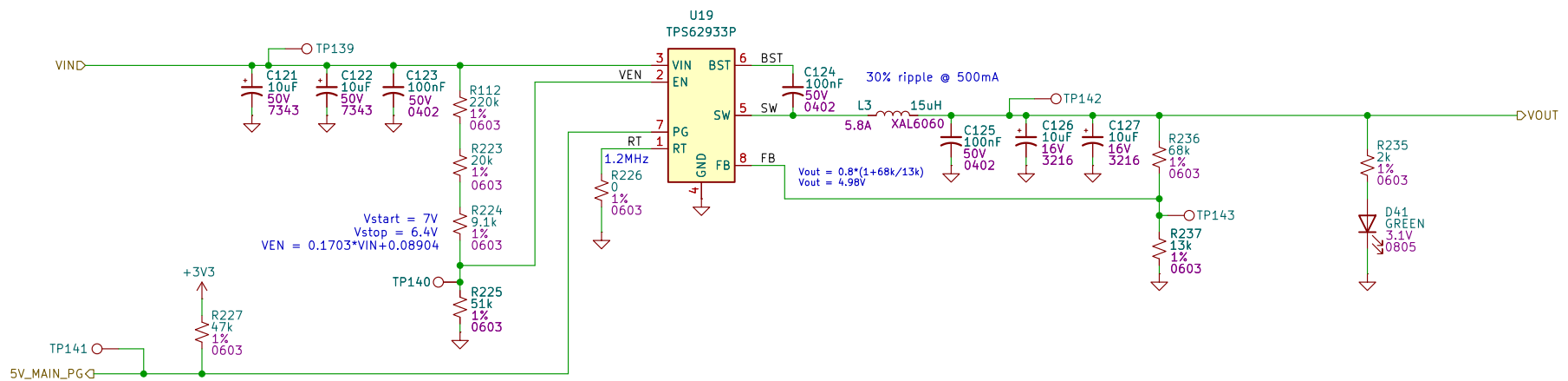


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Sheet: /System Power/
File: system_power.kicad_sch

Title: SYSTEM POWER

Size: A4	Date: 2026-08-27	Rev: A
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Sheet: /System Power/System 5V/
File: system_5V.kicad_sch

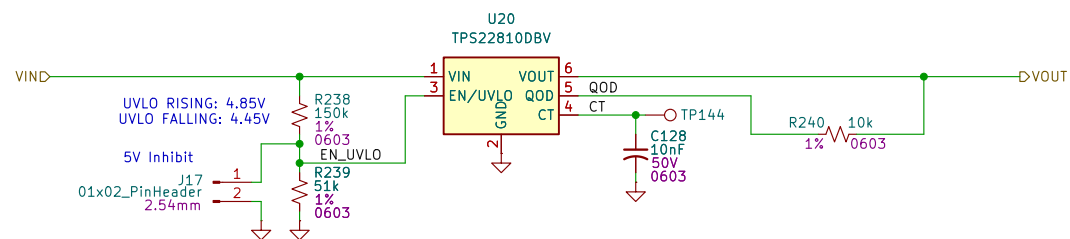
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Size: A4 Date: 2026-08-27

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Sheet: /System Power/LSW 5V/
File: lsw_5V.kicad_sch

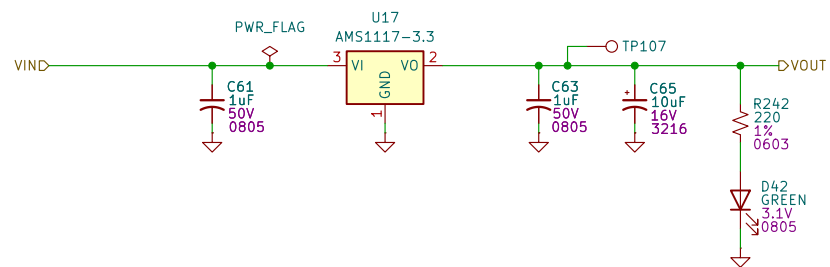
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Size: A4 Date: 2026-08-27

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Rev: A

Id: 5/32



Sheet: /System Power/LDO 3V3/
File: ldo_3V3.kicad_sch

Title: LDO 3V3

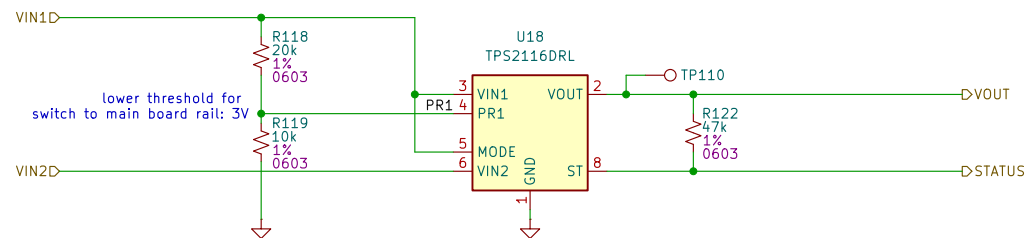
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Date: 2026-08-27

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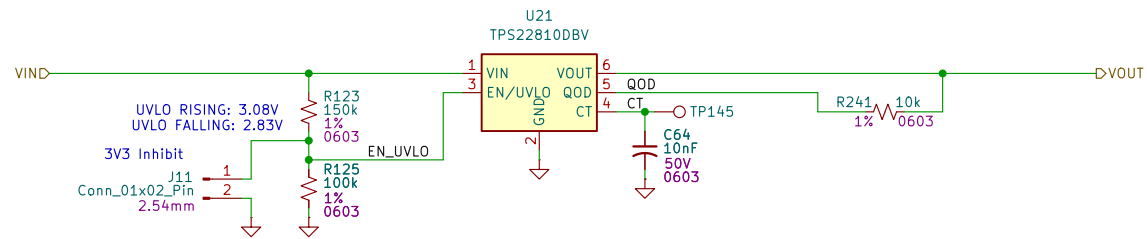
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Size: A4 Date: 2026-08-27

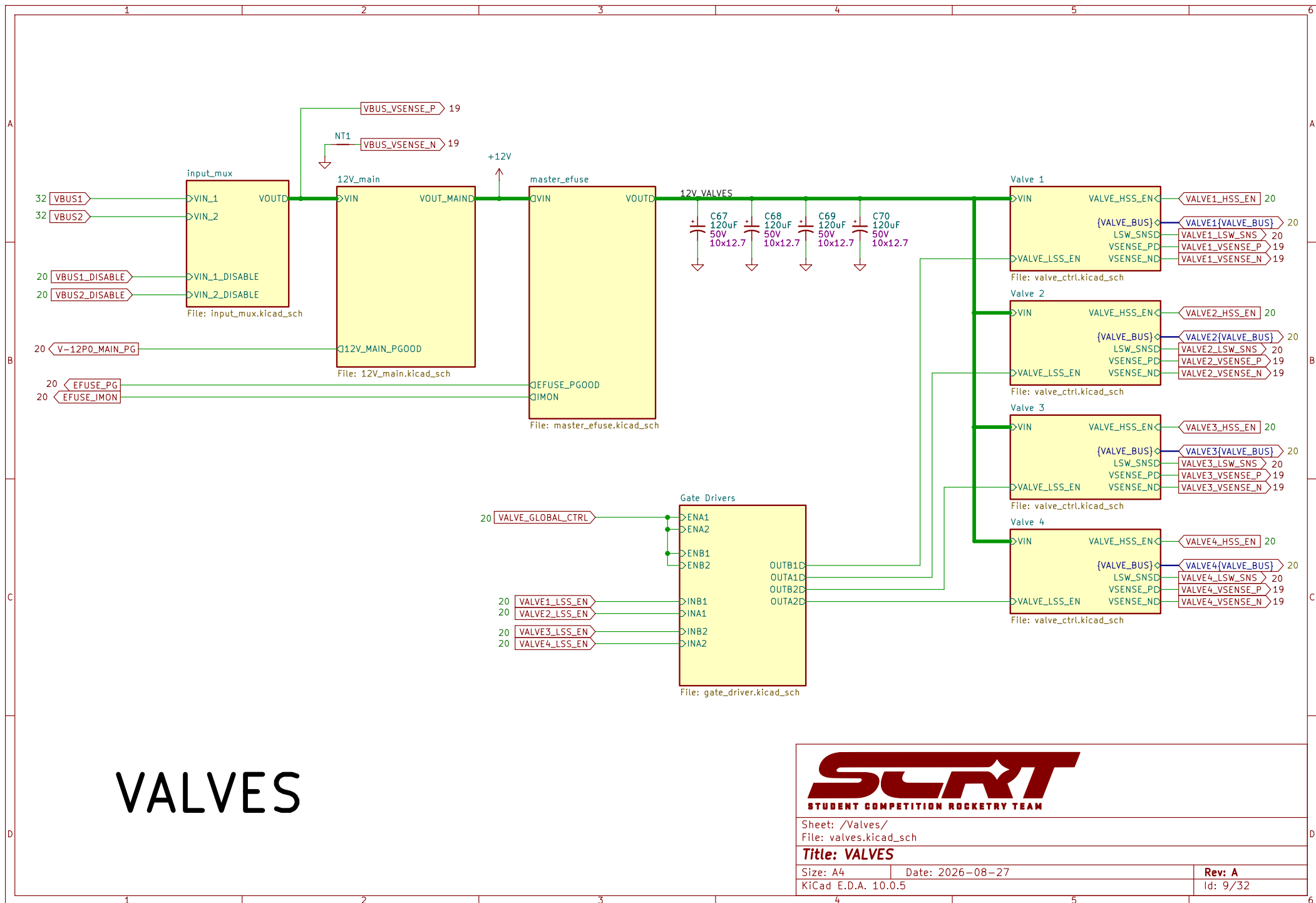
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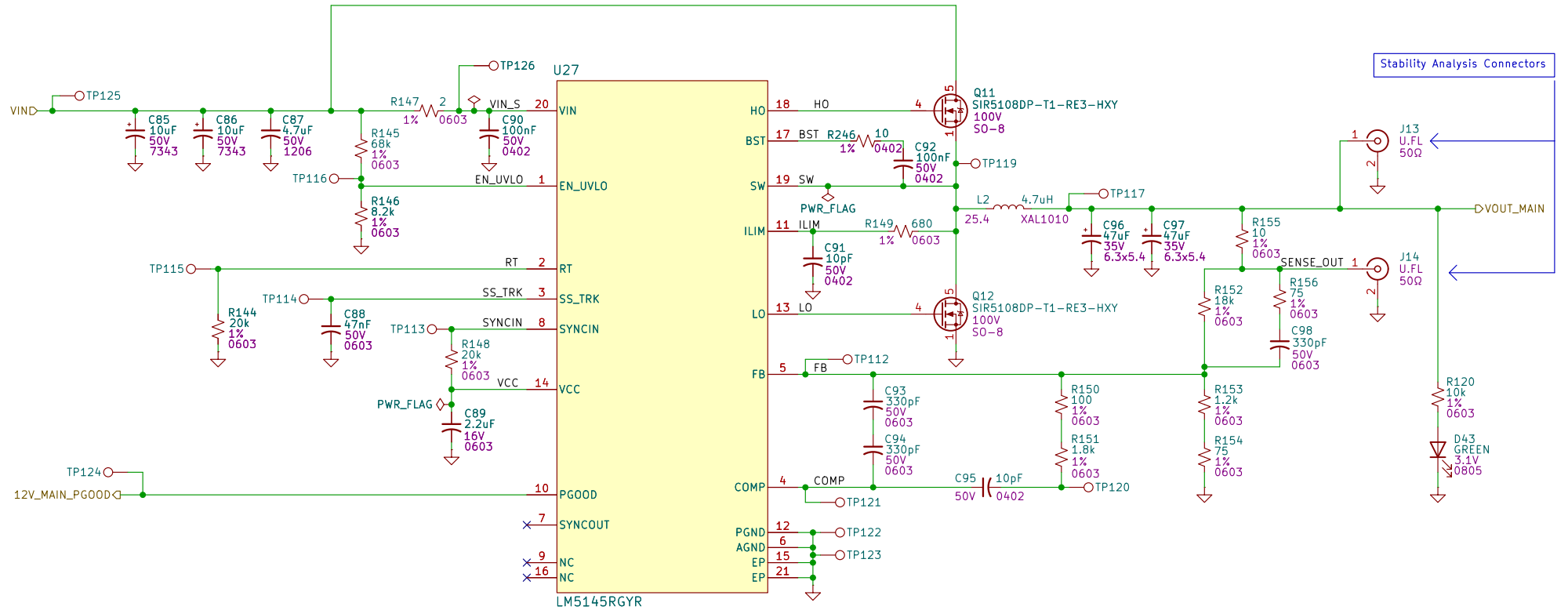
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Size: A4	Date: 2026-08-27	Rev: A
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12V MAIN



Sheet: /Valves/12V_main/
File: 12V_main.kicad_sch

Title: 12V MAIN

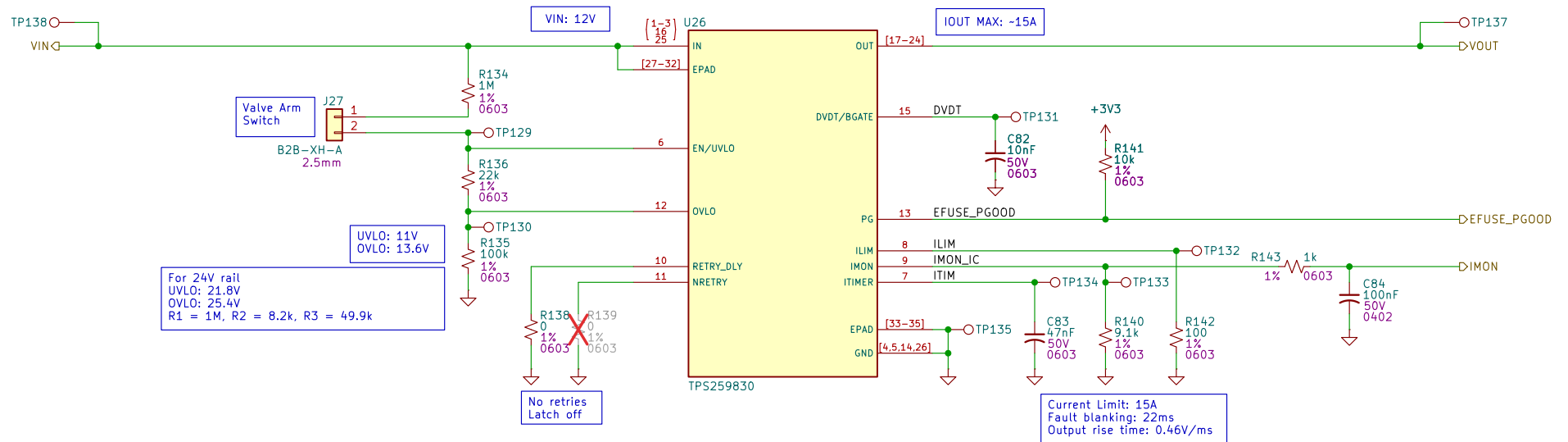
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EFUSE

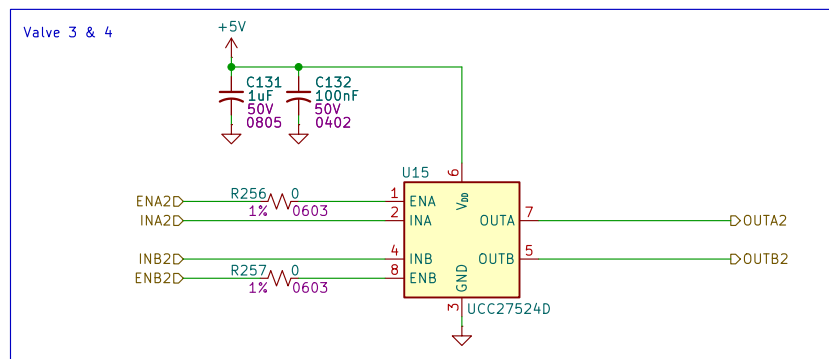
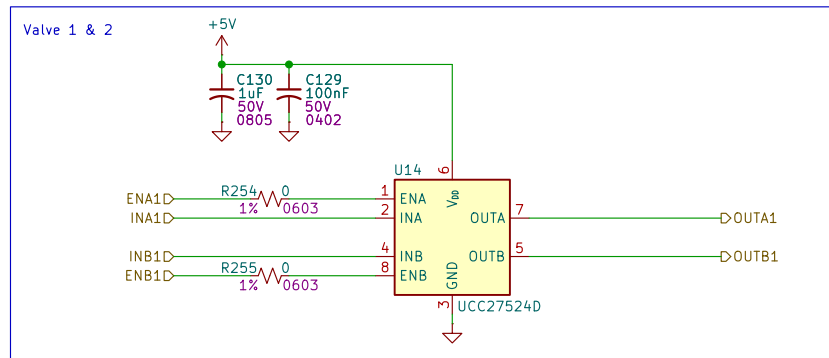


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File: master_efuse.kicad_sch

Title: EFUSE

Size: A4 Date: 2026-08-27
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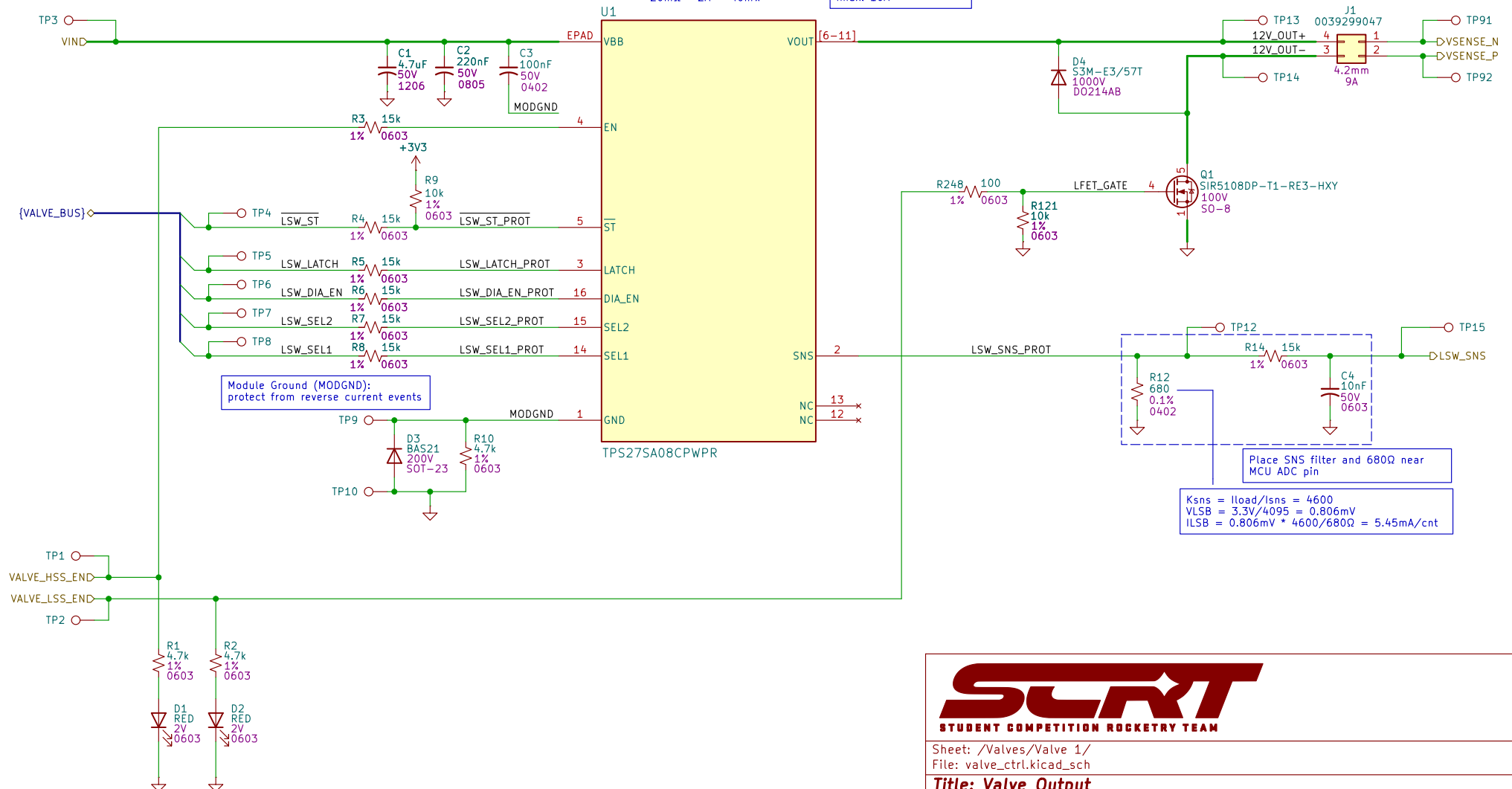
VALVE OUTPUT POWER

Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)

20mΩ * 2A = 40mW

12V, 2A nominal output
Vmax: 36V
Imax: 10A



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Sheet: /Valves/Valve 1/
File: valve_ctrl.kicad_sch

Title: Valve Output

Size: A4 Date: 2026-08-27

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Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)



File: valve_ctrl.kicad_sch

Title: Valve Output

Size: A4	Date: 2026-08-27
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Size: A4	
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Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)



Title: Valve Output

Title: Valve Output

Title: Valve Output

Size: A4	Date:
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Date: 2026-08-27

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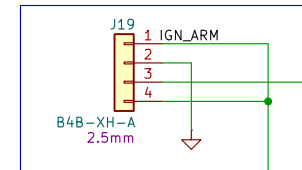
Rev: A

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IGNITER SYSTEM

ARMING SWITCH
EXTERNAL LED ARMED
CONNECTOR



D33 clamps Q8
Vsg at -15V
IOUT MAX: 14.7A

Continuity check:
1. ELOAD disabled, Q8 disarmed.
2. K1 shorts pins 1-3.
3. MCU checks continuity_sense net:
i. If HIGH, igniter not present
ii. If LOW, igniter present

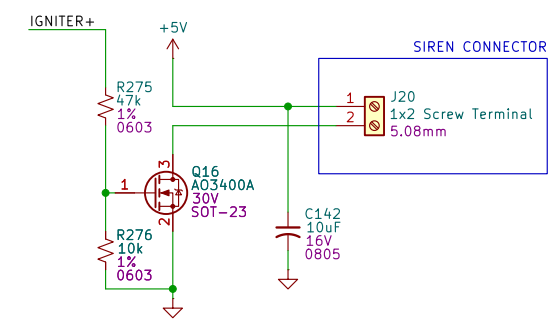
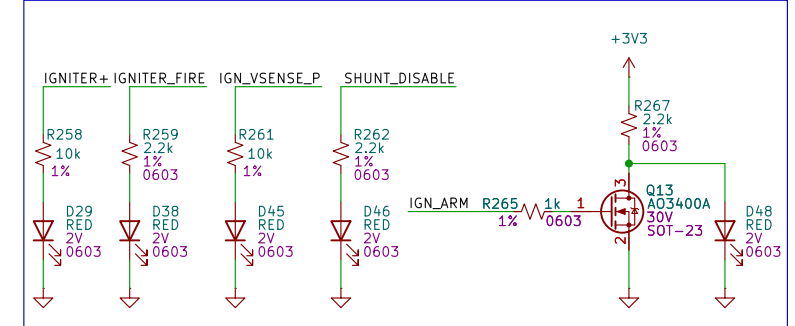
Current Limit: 14.7A
Fault blanking: 22ms
Output rise time: 0.46V/ms

sense @ 12V: 1.69V
sense @ 16V: 2.25V
sense @ 20V: 2.82V

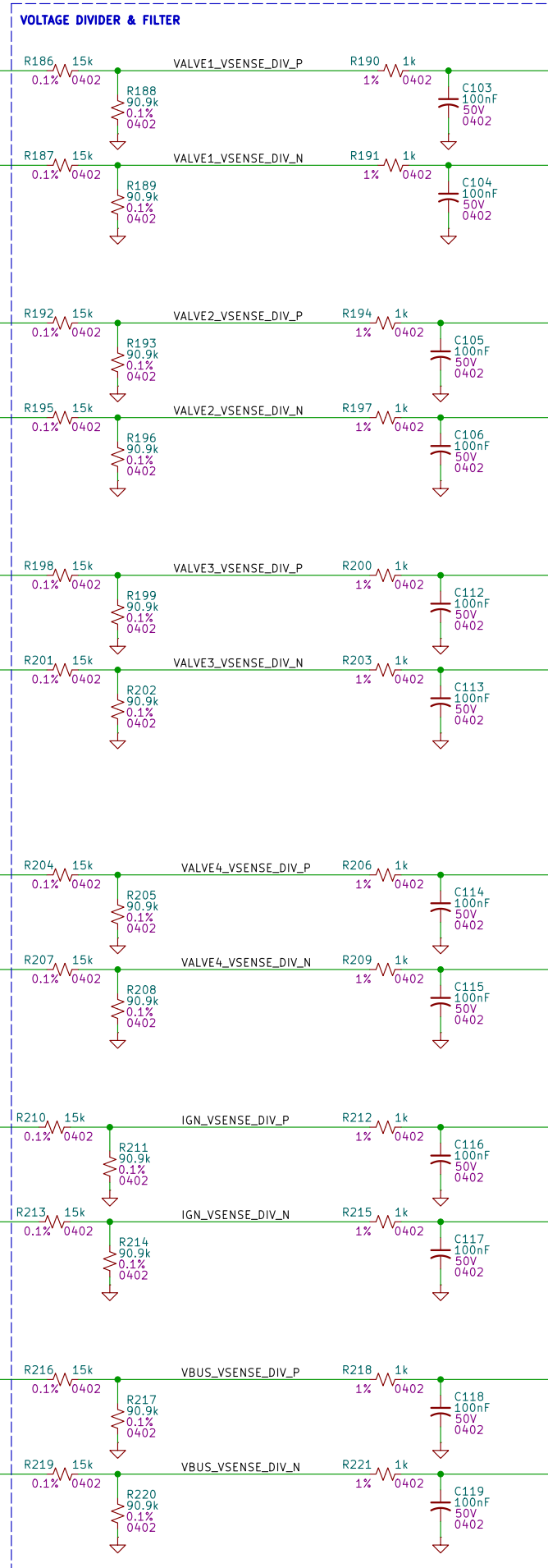
TEST POINTS

TP41 TestPoint	IGN_VSENSE_P	TP67 TestPoint	IMON
TP56 TestPoint	IGNITER_FIRE	TP68 TestPoint	ITIM
TP61 TestPoint	EFUSE_OVLO	TP69 TestPoint	SENSE
TP62 TestPoint	SHUNT_DISABLE	TP70 TestPoint	+5V
TP63 TestPoint	IGNITER+	TP71 TestPoint	IGN_RLY-
TP64 TestPoint	DVDT		
TP65 TestPoint	PGOOD_IGNITER		
TP66 TestPoint	ILIM		

LEDs



ADC



VALVE 1

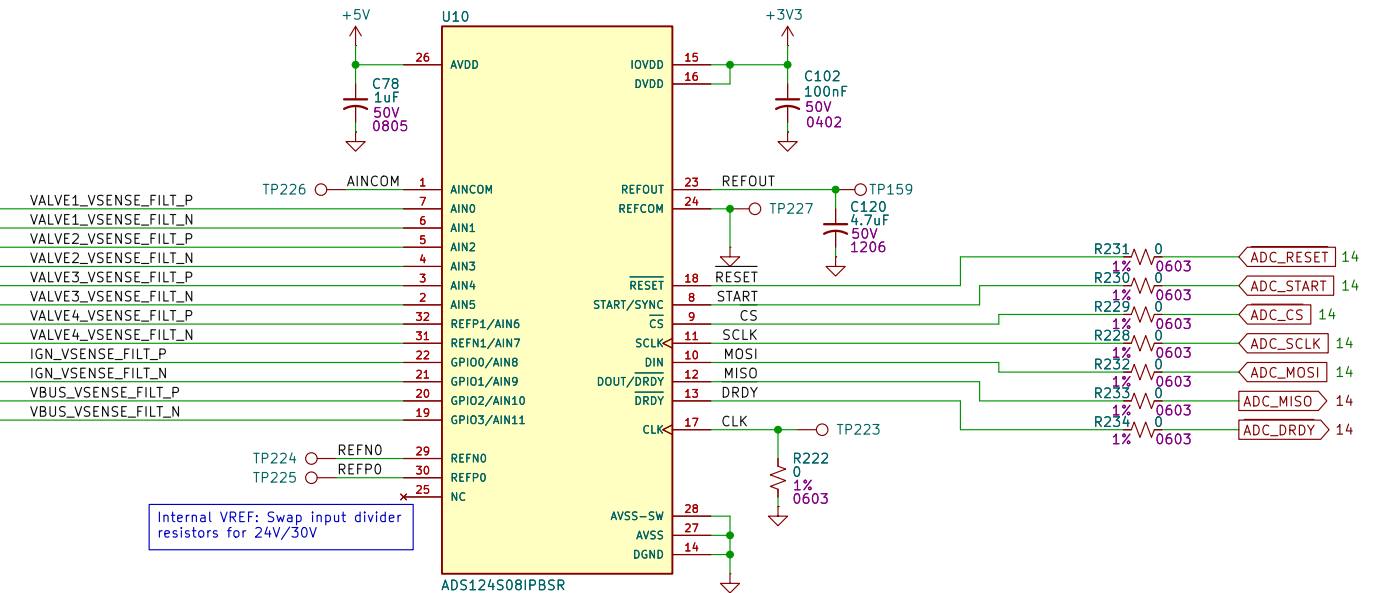
VALVE 2

VALVE 3

VALVE 4

IGNITER BATTERY INPUT

VALVE BATTERY INPUT

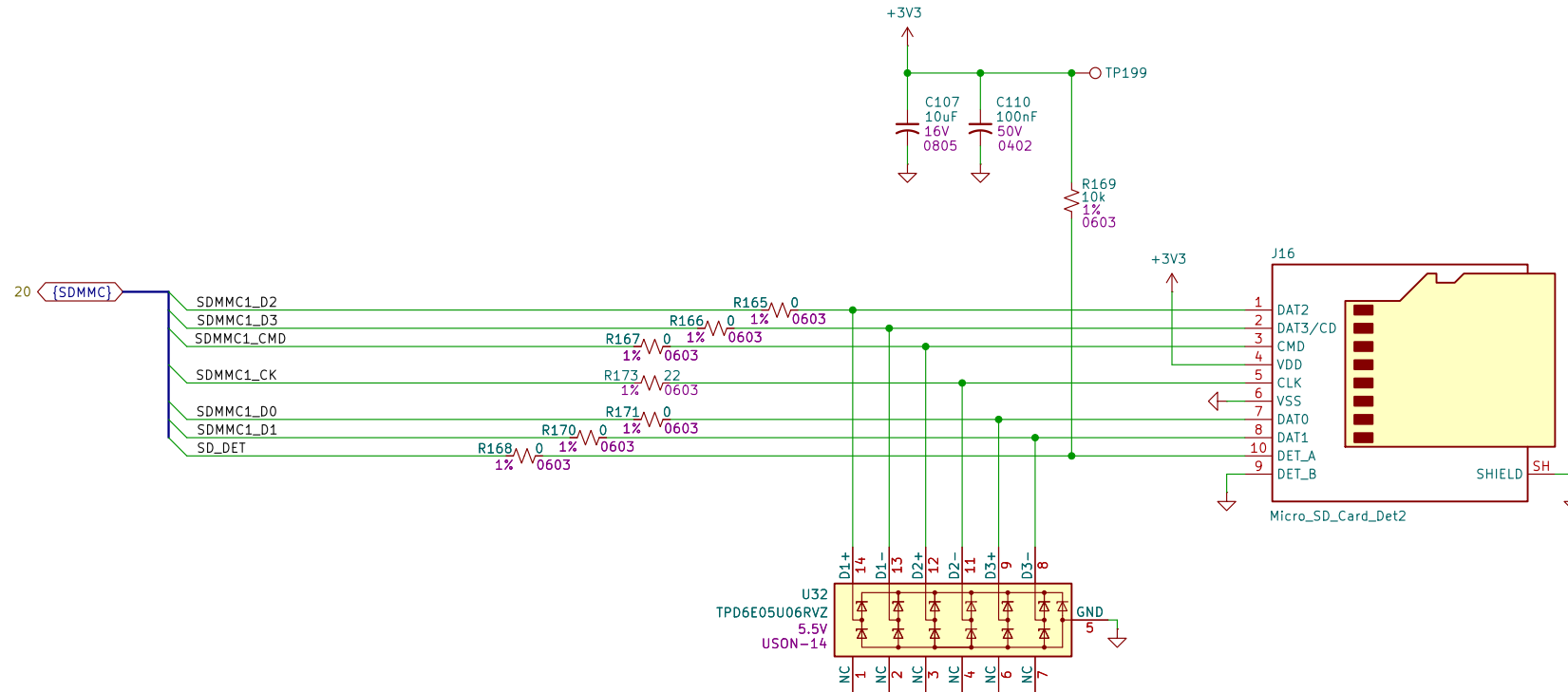


Internal VREF: Swap input divider resistors for 24V/30V



Sheet: /ADC/ File: adc.kicad_sch		
Title: ADC		
Size: A3	Date: 2026-08-27	Rev: A
KiCad E.D.A. 10.0.5		Id: 19/32

SD CARD



Sheet: /SD Card/
File: sd_card.kicad_sch

Title: SD CARD

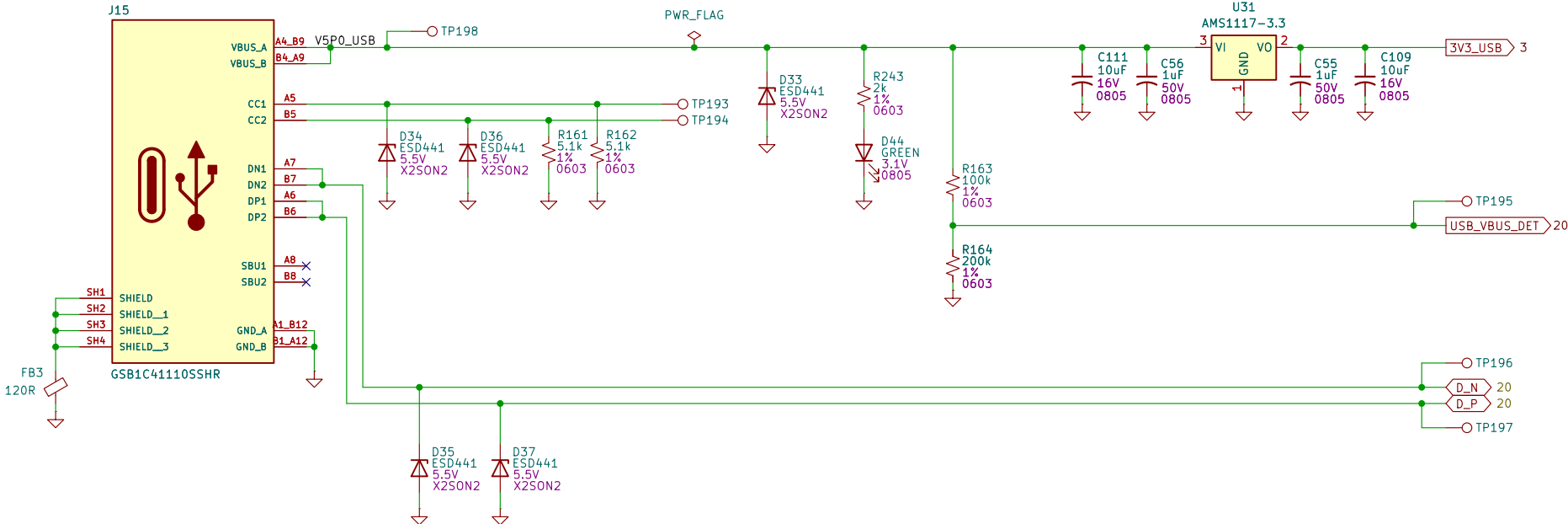
Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 21/32

USB-C



Sheet: /USB-C/
File: usb-c.kicad_sch

Title:

Size: A4

Date: 2026-08-27

Rev: A

KiCad E.D.A. 10.0.5

Id: 22/32

RS-485

Galvanic Power Isolation

The schematic diagram illustrates an RS-485 interface circuit with galvanic power isolation. The circuit is divided into several functional blocks:

- 3.3V Level Shifter:** U22 (74LVC1G17) shifts the 3.3V signal from the RS485_MAIN input to the RS485 transceiver.
- Power Regulation:** U23 (SN6505BDBVR) is a 420kHz switching regulator that converts +5V to a regulated 3.3V output. It includes a 420kHz switching inductor (TR1, 1:1.1) and a 420kHz switching capacitor (C108, 10uF, 16V, 0805). The output is filtered by C75 (1uF, 50V, 0805) and C54 (100nF, 50V, 0402).
- Galvanic Power Isolation:** U24 (ISO1450BDWR) is an isolated RS485 transceiver. It has two ground pins (GND1, GND2) and a PWR_FLAG pin. The output is connected to a 3-pin header (J12) via a 3mm pitch connector (0436500415).
- 5V ISO:** U25 (AMS1117-5.0) is a 5V LDO regulator that provides a clean 5V supply to the RS485 transceiver. It includes a 10uF, 16V, 0805 capacitor (C81) and a 10uF, 16V, 0805 capacitor (C80).
- Other Components:** The circuit includes various resistors (R129, R130, R131, R132, R133) and capacitors (C79, C51, C75, C54, C108, C81, C80) for filtering and timing.

Sheet: /RS-485/
File: rs-485.kicad_sch
Title: RS-485
Size: A4 Date: 2026-08-27 Rev: A
KiCad E.D.A. 10.0.5 Id: 23/32



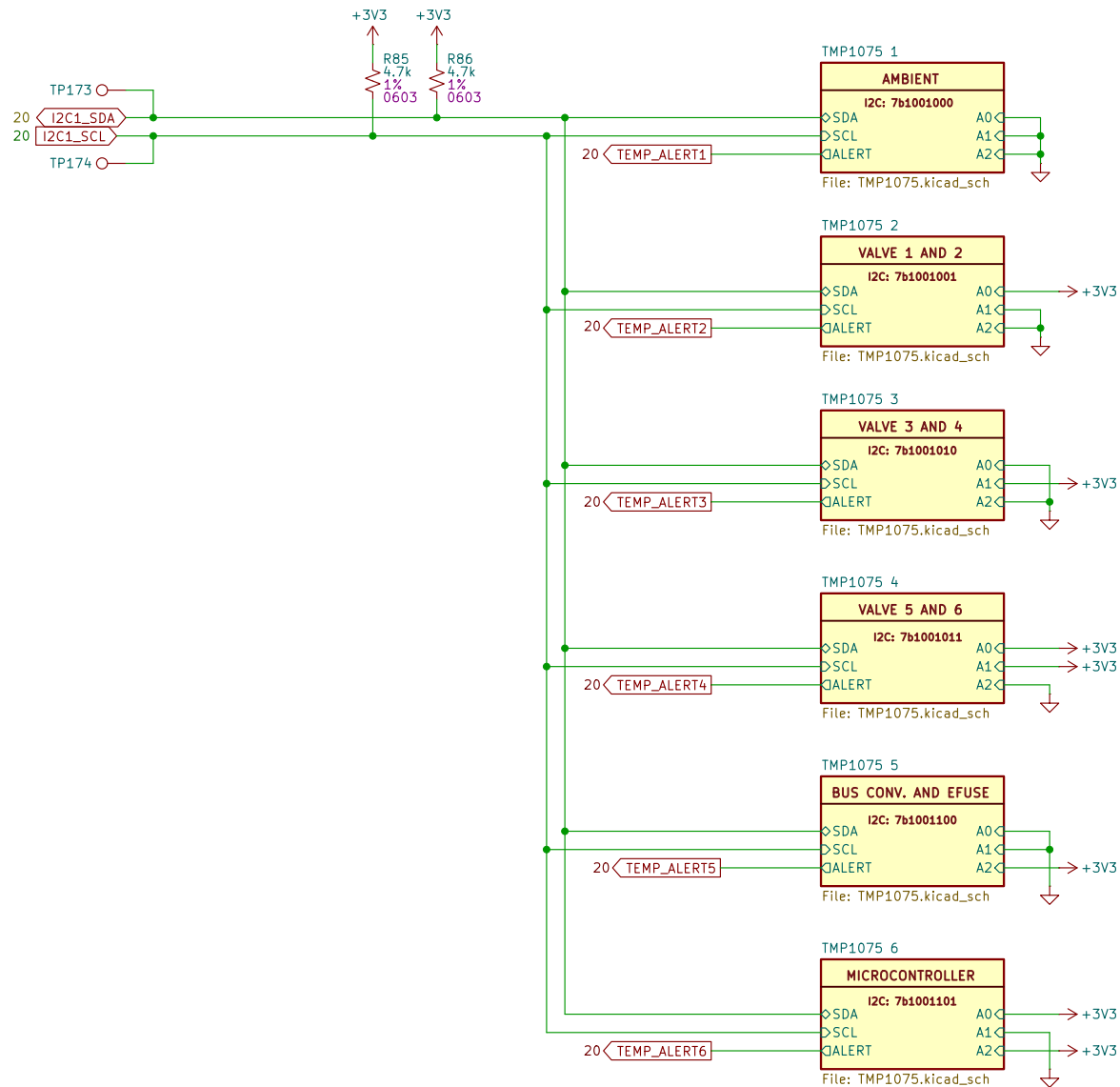
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Date: 2026-08-27

Rev: A

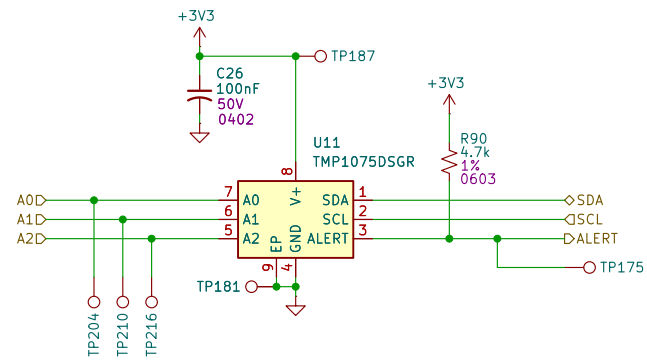
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TEMPERATURE SENSORS



 STUDENT COMPETITION ROCKETRY TEAM		
Sheet: /Sensors/ File: sensors.kicad_sch		
Title: SENSORS		
Size: A4	Date: 2026-08-27	Rev: A
KiCad E.D.A. 10.0.5		Id: 24/32

TMP1075



Sheet: /Sensors/TMP1075 1/
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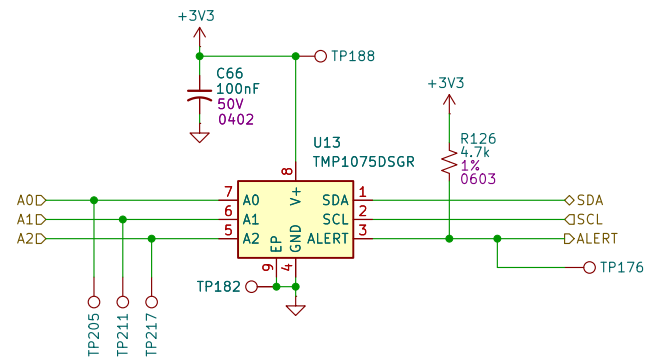
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Size: A4 Date: 2026-08-27

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Id: 25/32

TMP1075



Sheet: /Sensors/TMP1075 2/
File: TMP1075.kicad_sch

Title: TMP1075

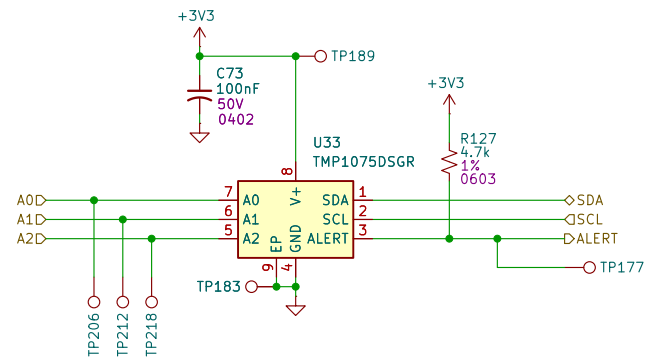
Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 26/32

TMP1075



Sheet: /Sensors/TMP1075 3/
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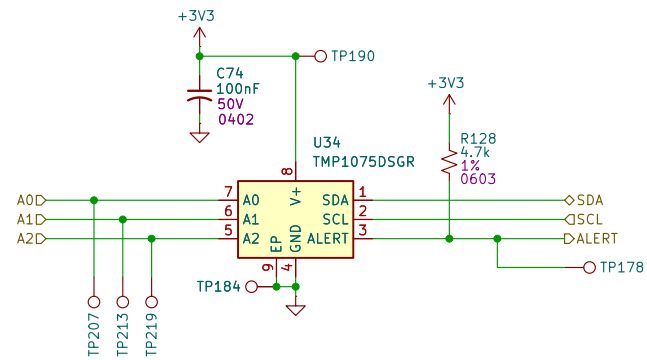
Title: TMP1075

Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A
Id: 27/32

TMP1075



Sheet: /Sensors/TMP1075 4/
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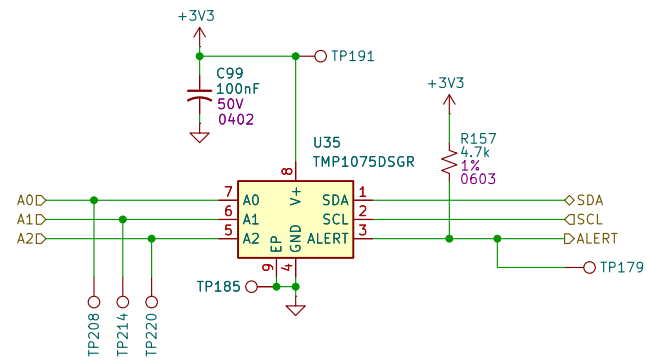
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KiCad E.D.A. 10.0.5

Rev: A

Id: 28/32

TMP1075



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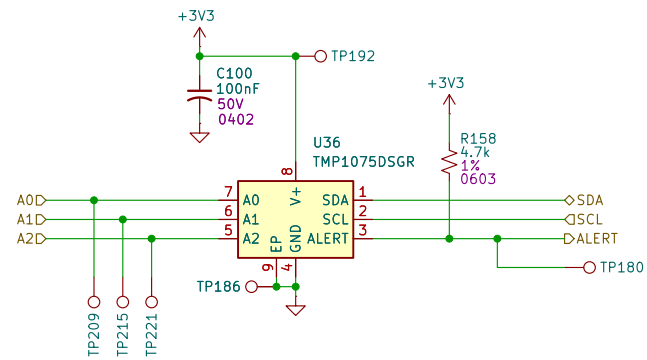
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KiCad E.D.A. 10.0.5

Rev: A

Id: 29/32

TMP1075



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Title: TMP1075

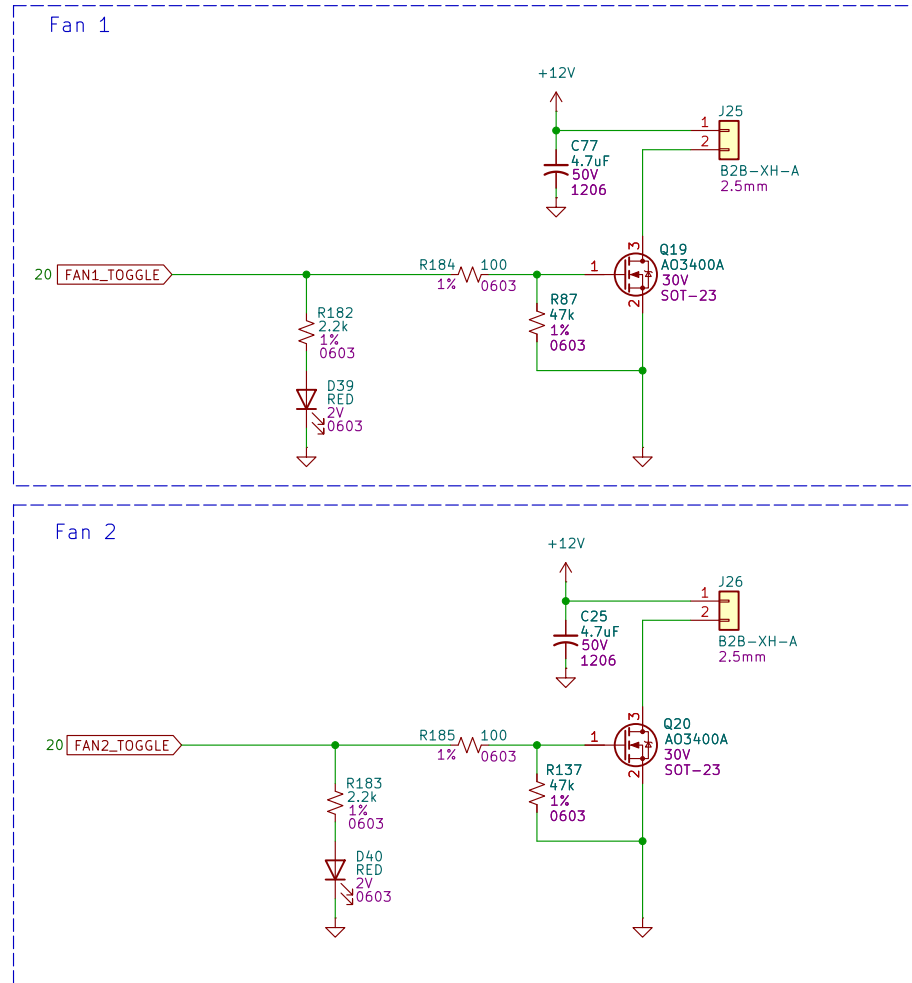
Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 30/32

FANS



Sheet: /Fans/ File: fans.kicad_sch		
Title: Fans		
Size: A4	Date: 2026-08-27	Rev: A
KiCad E.D.A. 10.0.5		Id: 31/32

A	
B	
C	
D	

